DEVI – IVR, SMS & WhatsApp A to Z Guide

Oct 27, 2026 · @Someone

1. Quick summary

DEVI SOS ku best combo: **India voice IVR ku Plivo (India-region account)**, **SMS ku India DLT route**, **WhatsApp ku direct Meta Cloud API** (already namma code la iruku). Global reach venumna Twilio voice + Meta WhatsApp.

SOS trigger aana guardian ku 3 channel la alert pogum:

1. **IVR call**: guardian phone ring aagum, TTS la emergency message pesum, "1 press pannunga" nu acknowledge (DTMF) vaangum.
2. **SMS**: internet illatha phone kum location link pogum.
3. **WhatsApp**: `devi_safety` template la name, location link, time pogum.

Option	Combo	Yaaruku best	Main risk
A (recommended, India-first)	Plivo voice + India DLT SMS + Meta WhatsApp	Guardians mostly India la	DLT registration mudiyama SMS/IVR live aagathu
B (global)	Twilio voice + SMS + Meta WhatsApp	Many countries, fast setup, Studio no-code	India call rate 4–5x costly, Indian number kidaikathu, international CLI la call pogum
C (developer control)	Telnyx voice + SMS + Meta WhatsApp	Low latency, full API control	India rate publish aagala, India numbers 4 cities mattum

Gold rule: **Plivo SMS/WhatsApp self-serve la illa** (Enterprise ~$1,000/month). So Plivo ah voice ku mattum use pannunga, SMS ah vera route la anupunga.

Intha document la iruka ella prices um October 2026 approx figures. Buy panna munnadi provider pricing page la verify pannanum.

2. Twilio, Plivo, Telnyx – yaaru ivanga?

Moonum **CPaaS (Communications Platform as a Service)** platforms. API/SDK vechu SMS anupalam, voice call (IVR menu) pannalam, WhatsApp message anupalam, pala countries la virtual numbers rent pannalam. Moonume "voice only" illa, Voice + SMS + WhatsApp ellam support pannum (Twilio la video, fax, email kooda iruku).

Provider	Founded / HQ	Core strength	Best known for
Twilio	2008, USA	Periya global coverage, best docs, Studio no-code flow builder	Enterprise reliability, athiga countries/number types
Plivo	2011, USA/India	Kammi per-minute/SMS cost, strong India presence	Budget voice & SMS, India–US corridor
Telnyx	2009, USA	Sontha private IP voice network (carrier resell illa)	Lowest latency voice, transparent pay-as-you-go, built-in AI voice tools

India-native providers (Exotel, Ozonetel, Knowlarity) um iruku. Avanga DoT/TRAI compliance ah base pannitu build pannavanga. India-only IVR ku avangalayum compare pannalam.

3. Global number vaanga mudiyuma? IVR panna mudiyuma?

Aamaa, moonu provider layum pala countries la number vaangalam. Every number (local, mobile, national, toll-free) ah IVR ku connect pannalam. Aana "global number" na ella country kum free ah call panra oru number illa:

Vishayam	Artham
Call receive panradhu	Ulagathula yaar venaalum call/SMS pannalam, caller oda carrier allow panna
Call out panradhu	Oru country ku dial panna account la andha country permission ON pannanum. Andha country oda per-minute rate pay pannanum
Number type	Ella country layum ella type (local/toll-free/mobile) kidaikathu. Sila country ku local ID/address proof venum

Provider	Number kidaikura countries	IVR markup	Notes
Twilio	100+	TwiML (`<Say>` , `<Gather>` , `<Dial>`) or Studio	Sila countries ku address proof, local contact, business registration venum
Plivo	65+	PlivoXML (TwiML mathiri)	Self-serve plan la US & India mattum. Mathathuku Enterprise
Telnyx	50+ (US/UK/EU strong)	TeXML (TwiML accept pannum) or Call Control API (JSON)	US/UK numbers cheap. Sila countries ku ID documents

Free global number kidaikuma? Illa. Ellame monthly rent. Trial credit mattum first number + konjam usage ah temporary ah cover pannum.

4. Free trial – actual ah enna kidaikum

Trial vechu namma sontha verified number la mattum dhan test panna mudiyum. Real guardians ku anupanumna payment + compliance mudikanum, moonu provider kum idhu dhan rule.

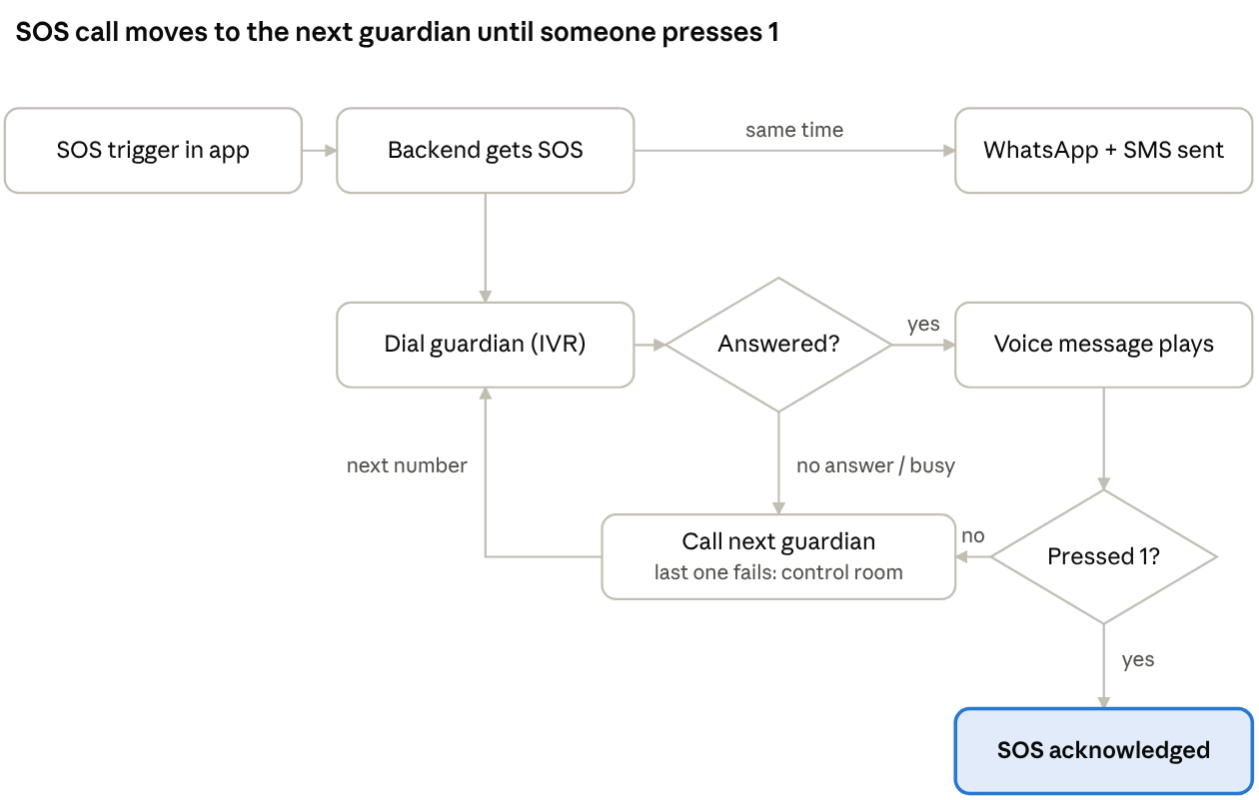
Provider	Free credit	Card venuma?	Restrictions
Twilio	~100 SMS, ~75 voice minutes, 3,000 SendGrid emails (fixed $ amount illa)	Start panna venaam	Console la verify panna numbers ku mattum call/SMS. 1 active trial number (lifetime max 3). Live ku card add panni upgrade
Plivo	$10 credit	Start panna venaam	Self-serve usage $2,500/month cap. US & India mattum. SMS, WhatsApp, Verify API self-serve la illa (Enterprise + sales)
Telnyx	Pretrial $25 AI credit (card illama); verified trial ku $5 testing credit	Pretrial ku venaam; full sending ku card venum	Business email mandatory (Gmail/Yahoo reject). Scale panna KYC document check

WhatsApp testing: Twilio trial venaam. Meta developer console (developers.facebook.com → WhatsApp → API Setup) la free test number kudupaanga. Max 5 recipient numbers add panni OTP vechu verify pannitu test pannalam.

5. IVR capability & DEVI SOS call flow

Moonu provider layum Voice IVR + SMS + WhatsApp ore account la pannalam. IVR na namma server oru XML-mathiri markup return pannum, provider adha pesi key press vaangum.

Provider	IVR markup	No-code option	SMS/WhatsApp kooda?
Twilio	TwiML	Twilio Studio (drag & drop)	Aamaa, same account
Plivo	PlivoXML	Plivo Flow (basic)	Aamaa, aana Enterprise plan venum
Telnyx	TeXML (TwiML-compatible) or Call Control API	Limited, code-first + AI voice builder	Aamaa



DEVI SOS IVR flow · 2 decisions, next-guardian loop

WhatsApp + SMS um IVR call um ore nerathula pogum. Call la yaarum 1 press pannala na adutha guardian ku pogum; ellarum fail aana control room ku (`ADMIN_WHATSAPP`) pogum.

Step by step:

1. App la SOS press → backend `POST /api/sos/trigger` .
2. Backend WhatsApp (`devi_safety` template) + SMS anupum, same time la provider ku outbound call request (`calls.create`) anupum.
3. Provider guardian number ku PSTN la dial pannum.
4. Guardian edutha, provider namma webhook ku `POST` pannum (e.g. `/api/ivr/answer`).
5. Backend XML return pannum: `<Say>` emergency message + `<Gather numDigits="1">` "1 press pannunga".
6. Guardian 1 press → provider `/api/ivr/gather` ku digit anupum → SOS "acknowledged" nu dashboard la update.
7. Eduthala / busy / voicemail → status callback (`/api/ivr/status`) → adutha guardian ku call.

Voicemail problem: voicemail call ah "answer" pannidum. Answering Machine Detection (AMD) on pannunga, illana DTMF "1" vandha mattum dhan acknowledged nu count pannunga.

Inbound helpline – guardian thirumbi call panna

Guardian missed call ah paathu adhe number ku thirumbi call pannuvaanga, so **SOS call panna adhe Plivo number la inbound IVR venum.** Illana call dead number ku pogum. Thaniya number venaam, cost ~₹0.38/min dhan.

Flow:

1. Guardian Plivo 080/022 number ku call → Plivo namma `/api/ivr/inbound` webhook ah call pannum (caller number `From` la varum).
2. Backend `From` number ah active SOS la registered guardian ah nu check pannum.
3. **Registered guardian + active SOS iruntha:** "Ungaloda contact ku SOS active ah iruku. Details WhatsApp/SMS la anupirukom. Control room kooda pesa 2 press pannunga." Indha call ah acknowledged ah mark pannalam (guardian respond pannaanga).
4. **2 press panna:** `<Dial>` vechu control room phone ku connect.
5. **Theriyaadha number or active SOS illa:** general message mattum ("This is the DEVI safety line"). Location, name edhuvum solla koodathu.

Item	Detail
Number	Adhe Plivo India number (outbound caller ID)
Inbound cost	₹0.38/min (Plivo India)
Control room ku forward	Inbound ₹0.38 + outbound leg ₹0.38 = ~₹0.76/min
Toll-free (1800) venumna	Plivo toll-free inbound ~₹1.30/min (conflicting listings, confirm), Knowlarity toll-free ~₹1.05/min
Security	Caller ID spoof aagalam, so location/name voice la solla koodathu; already WhatsApp/SMS la secure ah poyirukum
Code	`ivr.routes.js` la `/inbound` route + signature validation (Section 13.3 mathiri)

112 (ERSS) kooda integrate pannalama?

Ippo private apps ku 112 ERSS public API illa; direct integration mudiyathu. ERSS MHA keezha Nirbhaya Fund la run aagudhu, C-DAC build pannadhu; ovvoru state kum Emergency Response Centre (ERC) iruku.

Option	Eppadi	DEVI ku
App la "Call 112" button	SOS screen la one-tap `tel:112`	Ippove pannalam, cost illa. Best quick win
User ku 112 India app suggest	Official app la SOS, location state control room ku pogum; SHOUT feature women ku volunteers alert	Onboarding la recommend pannalam
Formal integration	State ERC / state police or MHA / C-DAC kitta partnership	Long process, public path illa; partnership ah try pannalam

Presentation answer: "DEVI 112 ah replace pannala. Guardian network + control room ah alert pannudhu; app la 112 one-tap button um iruku. Official integration ku state ERC kitta pesanum."

Sources: 112.gov.in, PIB on ERSS, ERSS explainer (Factly).

Guardian unknown number call ah edukkalaina?

Theriyaadha 080 number la irundhu vara call ah pala per edukka maataanga, idhu dhan IVR oda periya real-world risk. Naalu vazhi la fix pannalam:

1. **Guardian add pannumbodhu:** "DEVI SOS line" number ah contact ah save panna sollunga (T2 SMS + WhatsApp la vCard/number anupunga).
2. **Same number always:** oru fixed Plivo number dhan use pannunga; maathita save panna contact waste.
3. **Truecaller Verified Business Caller ID:** green badge + business name kaatum, pick-up rate increase aagum. India pricing public ah illa; Truecaller business team or Route Mobile mathiri partners kitta quote. Verified badge irundhalum users spam mark panna mudiyum.
4. **WhatsApp + SMS first:** call mattum rely pannama, rendu message um call ku munnadi / kooda pogum (already flow la iruku).

Sources: Truecaller for Business (Capterra India), Route Mobile verified caller ID webinar.

6. Pricing – number, voice, SMS, WhatsApp, MMS

India calls ku Plivo dhan cheap (₹0.38/min ≈ $0.0045); Twilio India ku ~$0.05–$0.07/min, 5–10x costly. Ella figures um approx USD, October 2026.

6.1 Monthly number rental

Number type	Twilio	Plivo	Telnyx
US local	$1.15/mo	$0.50/mo	$1.00/mo
US toll-free	$2.15/mo	$1.00/mo	~$1/mo (800 numbers: $500 one-time/yr, apram $40/mo)
UK local / mobile	$3.50 / $2.50 per mo	Country poruthu	Country poruthu
India	Kidaikathu	₹200/mo (about $2.40)	Mobile numbers 4 cities mattum, rate confirm aagala
Other (e.g. Nigeria)	Often $1–$5/mo	Country poruthu	Nigeria $25/mo la irundhu

6.2 Voice per minute

Direction	Twilio	Plivo	Telnyx
US outbound	$0.014	$0.010–$0.012	Competitive (live rate card)
US inbound to local	$0.0085	$0.0055–$0.0085	Similar
US inbound to toll-free	$0.022	Twilio vida konjam kammi	Competitive
India outbound (local / mobile)	$0.0699 / $0.0496	₹0.38 in & out, 30-sec pulse (official India page)	Public page la India rate illa (Oct 2026 check panninom). Base $0.002/min API + destination SIP rate; India rate Mission Control rate deck or sales quote la dhan
Call recording	$0.0025	India free (90 days storage)	Live rate check

India la irundhu vera country ku call: "international flat rate" illa. Destination country oda rate dhan (e.g. US ku Twilio ~$0.014/min).

6.3 SMS & WhatsApp (provider fee mattum)

Item	Twilio	Plivo	Telnyx
SMS US outbound	~$0.0079/msg + carrier fee	Twilio vida kammi (self-serve US/India mattum)	Competitive
WhatsApp provider fee	$0.005/msg (sent & received)	$0.0044/msg (Enterprise plan la)	$0.004/msg
WhatsApp calling	Supported	$0.0040/min	$0.0025/min (in & out)

Meta charge thaniya: July 2025 la irundhu Meta per message bill pannum (24-hour conversation illa). Oct 1, 2026 la irundhu per phone number per month first 1,000 *service* messages free. SOS alert business-initiated utility template, so adhu free list la varaathu. Oct 1, 2026 la irundhu customer service window kulla anupura utility messages um charge aagum (munnadi free). India utility template approx ~₹0.12/message (Meta rate card la verify pannunga). **Direct Meta Cloud API use panna provider fee zero** – DEVI already ippadi dhan pannudhu.

6.4 MMS (picture/video message)

Provider	Outbound MMS	Inbound MMS
Twilio	~$0.022/msg (US) + carrier fee ~$0.007–$0.01	~$0.01–$0.02/msg
Plivo	US numbers la support, rate publish aagala (sales kitta kelunga)	Same
Telnyx	$0.015/part (US) + carrier fee	$0.005/part (US) + carrier fee

India la MMS practically use illa; WhatsApp dhan media ku.

6.5 Cost examples

Twilio IVR, US, 100,000 inbound + 50,000 outbound min/month: inbound 100,000 × $0.0085 = $850, outbound 50,000 × $0.014 = $700, recording 150,000 × $0.0025 = $375. Total **~$1,925/month** + number rent. Plivo/Telnyx usually 20–40% kammi.

DEVI India voice, 5,000 SOS/month, oru SOS ku 2 guardian × 1 min = 10,000 min:

Provider	Calculation	Voice cost/month
Plivo	10,000 × ₹0.38	₹3,800 (about $45)
Twilio	10,000 × $0.0496–$0.0699	~$496–$699

Idhoda SMS (DLT route rate provider poruthu) + WhatsApp (~₹0.12 × messages) + number rent add pannanum. Average call length and guardians per SOS maarina cost um maarum.

6.6 India SMS rates (DLT route)

India transactional/service SMS ku **~₹0.15–₹0.25 per SMS** dhan market rate; Twilio India ~₹0.45, idhu 2–3x costly. Ella providers um same TRAI DLT route use pannuvaanga, so rates close ah irukum; volume and hidden fees dhan difference.

Provider	Per SMS (approx, GST thaniya)	Notes
MSG91	₹0.25 (5,000 SMS pack) → ₹0.16 (~9.6 lakh pack); negotiated ₹0.13 varai	Prepaid packs
Gupshup	₹0.10–₹0.18 transactional	Public rate card illa; wallet minimum ₹10,000–₹25,000 nu reports
Exotel	₹0.18 la irundhu	Exotel plan kooda add-on; sila reports 1 lakh minimum volume
Twilio (India)	~₹0.45 (forex surcharge kooda)	DLT idhukum venum

Hidden costs: sticker rate mela ~25–40% add aagalam (DLT scrubbing fee, failed-delivery buffer, wallet minimum) + 18% GST. Quote vaangum bodhu "all-in rate" kelunga.

Sources: MSG91 pricing (aggregator), Gupshup pricing audit, SMS OTP pricing India, Exotel pricing guide. Ellam third-party; final quote provider sales kitta.

6.7 Monthly total cost – ella channel um sethu

5,000 SOS/month ku Plivo combo ~₹6,750–₹7,650 (oru SOS ku ~₹1.5); Twilio combo ~₹47,450–₹64,450, 7–8x costly. Voice dhan periya cost; SMS + WhatsApp chinna part.

Assumptions: 5,000 SOS/month, oru SOS ku 2 guardians, guardian ku 1 call (avg 1 min) + 1 SMS + 1 WhatsApp. So 10,000 call-min + 10,000 SMS + 10,000 WhatsApp. $1 = ₹84.

Item	A: Plivo + MSG91 + Meta (recommended)	B: Twilio voice + SMS + Meta	C: Exotel voice + SMS + Meta
Voice (10,000 min)	₹0.38/min → ₹3,800	$0.0496–$0.0699/min → ₹41,700–₹58,700	₹0.80–₹1.15/min (historical) → ₹8,000–₹11,500
Number rent	₹200	US number ~₹100 (Indian number illa)	Quote venum
Plan / platform fee	Illa (pay-as-you-go)	Illa	Believer ₹19,999 / 12 months (usage credit kooda) → ~₹1,700/month
SMS (10,000)	₹0.16–₹0.25 → ₹1,600–₹2,500	~₹0.45 → ₹4,500	₹0.18 → ₹1,800
WhatsApp (10,000, direct Meta)	~₹0.115 → ₹1,150	₹1,150	₹1,150
AMD, recording, Tamil/Hindi TTS	Free (Plivo India page)	Extra (recording $0.0025/min)	Quote
Monthly total (GST thaniya)	**~₹6,750–₹7,650**	**~₹47,450–₹64,450**	**~₹12,650–₹16,150 + number**
Oru SOS ku	~₹1.35–₹1.55	~₹9.5–₹12.9	~₹2.5–₹3.2

One-time: DLT ~₹5,900; Plivo KYC, Meta verification free. **GST 18%** ellathukum mela.

Scale: cost SOS count kooda linear ah maarum (e.g. 1,000 SOS/month na A ~₹1,500). Call 30 sec la mudinja Plivo 30-second pulse la half cost dhan.

Sources: Plivo India voice pricing (₹0.38/min in & out, ₹200/month number, 30-sec pulse), Meta WhatsApp pricing, WhatsApp India rates (Robylon), Exotel pricing guide. Twilio rates Section 6.2 la irundhu.

7. India – certificates, DLT, numbers, rates

Aamaa, India number ku automated IVR call/SMS anupa registration venum. TRAI oda TCCCPR rules (2024–2025 amendments) apply aagum. DLT mudiyama SMS operator level la drop aagum.

7.1 TRAI rules

Requirement	Artham
DLT registration	Business ah "Principal Entity" (PE) ah operator DLT portal la register pannanum, apram Registered Telemarketer (RTM) kooda link. Usually CPaaS provider idha arrange pannuvaanga
Number series	160-series = BFSI service/transactional calls. 140-series = promotional. Non-BFSI service calls (DEVI) Plivo 080/022 number la pannalam
DND / NCPR	Promotional calls Do-Not-Disturb registry numbers ku poga koodathu
Calling window	Promotional calls 9 AM – 9 PM IST mattum
Consent	Explicit consent venum. Violation: 1st time 15-day service bar, repeat panna 1-year blacklist

DEVI SOS = **transactional / service** category. Promotional illa, so time window restriction illa. Aana guardian consent record pannanum.

7.2 DLT registration steps

1. **Entity documents ready:** Company PAN, GST certificate or Certificate of Incorporation (or Udyam), company letterhead la Letter of Authorization, authorized signatory ID/address proof.
2. **PE registration** oru operator DLT portal la (Jio TrueConnect, Airtel, Vi). Approx ₹5,900 one-time, usually 1–3 working days (sila case 7 varai). Full details Section 12 la.
3. **Header (Sender ID)** register: 6 characters, e.g. `DEVIAP`. 1–2 days.
4. **Content templates** register: SMS text + variables `{#var#}`. Anupura text template kooda exact ah match aaganum, illana drop. 24–48 hrs.
5. **Consent template** (transactional/service ku venumna).
6. PE ID, Header, Template ID ah CPaaS account la bind pannunga → India live ready.

7.3 Indian virtual number vaanga documents

Provider	Enna venum
Plivo	Dedicated India-region account + GST Certificate / Certificate of Incorporation / Udyam Registration la onnu (PAN mattum accept illa). First application ku authorized signatory signature + company seal
Telnyx	Mobile numbers 4 cities mattum (Mumbai, Gurugram, Noida, Bangalore). Andha city la business/residential address proof + KYC
Twilio	Standard Indian local number kudukathu. India ku outbound call international (non-Indian) number la irundhu mattum. India ku Voice OTP/2FA blocked

Twilio CLI risk: Twilio call international caller ID la varum. Indian operators international-CLI calls ah label/block pannalam, guardian edukama poga chance iruku. SOS ku idhu periya risk.

OSP licence: DoT 2020–21 la mandatory OSP licence ah eduthutaanga. Providers ippo VNO authorisation la operate pannuvaanga. Rules adikadi maarum; production ku munnadi DoT or telecom lawyer kitta confirm pannunga.

7.4 India-first project ku practical note

Twilio India ku best fit illa: Indian number illa, Voice OTP block, highest India rate. Plivo (India-region account) India-friendly. Telnyx India expand pannudhu (4 cities), aana VNO scope ah support kitta confirm pannunga. India-only ku Exotel, Ozonetel, Knowlarity kooda compare pannunga.

7.5 Exotel / Ozonetel / Knowlarity comparison

DEVI SOS mathiri automated outbound IVR ku Plivo dhan simple & cheap. Indha moonu India-native providers um contact-centre (agents, plans, per-user fee) ku build pannadhu; pure API pay-as-you-go illa. Public rate card yaarukum illa, so ellam third-party figures.

Provider	Pricing model	Plans (approx)	Per-minute (approx)	SMS	DEVI ku fit
Plivo (compare ku)	Pay-as-you-go API	Illa	₹0.38/min in & out (official page)	Self-serve la illa	Best: API-first, AMD + TTS free

Provider	Pricing model	Plans (approx)	Per-minute (approx)	SMS	DEVI ku fit
Exotel	Prepaid plan (software rental + usage credit)	Micro ₹5,555/3 mo, Dabbler ₹9,999/6 mo, Believer ₹19,999/12 mo, Influencer ₹49,499/12 mo	Historical: local out ₹0.80, national out ₹1.15, local in ₹0.40	₹0.18 la irundhu	Good backup: API iruku, DLT/TRAI handle pannuvaanga; plan commitment venum
Ozonetel	Per user per month	Starter $25, Standard $45, Premium $55 per user/month, Enterprise custom	Publish aagala (subscription ku mela thaniya)	Thaniya	Low: agent-based call centre ku dhan
Knowlarity	Per agent + per-minute/pulse	₹1,999–₹3,499 per agent/month; Advance ₹16,800/year (1,200 free min/month)	30 paise / 30 sec; toll-free incoming ₹1.05/min	Thaniya	Low–medium: toll-free helpline ku consider pannalam

GST 18% ellathukum mela. Final rate sales quote la dhan confirm aagum.

Sources: Exotel pricing guide (DialNexa), Exotel pricing (CloudTalk), Ozonetel pricing (CloudTalk, Jul 2026), Knowlarity IVR (FitGap), CCaaS pricing India 2026.

7.6 TRAI 2025 amendment & 1600-series – DEVI ku impact

Ippo DEVI ku 1600-series kattayam illa. 1600-series DoT Government + BFSI (RBI, SEBI, PFRDA, IRDAI regulated) ku mattum dhan allocate pannirukku; adhanala non-BFSI ku andha number vaanga mudiyathu. Aana TRAI phases extend aagura chance iruku, adhu dhan risk.

Point	Detail
TCCCPR 2nd amendment	12 Feb 2025 publish; category + consent rules 14 Mar 2025 la irundhu
1600-series direction	TRAI 19 Nov 2025: BFSI service/transactional calls 1600 la irundhu mattum. Deadlines 1 Jan 2026 (banks) to 15 Mar 2026 (stockbrokers)
140-series	Promotional calls mattum
Non-BFSI service calls	Clear series rule illa. Plivo 080/022 number non-BFSI service/transactional calls ku allow pannudhu

Point	Detail
Consent (vendor reports)	Explicit consent record (OTP or digital) venum; sila guides "service explicit" rules tight aagudhu nu solludhu

DEVI position (presentation la solla): SOS call user trigger panna personal emergency relay; marketing illa. Guardian ah add pannumbodhu OTP vechu consent vaangi record pannrom. Adhanala service category la varum.

Action items:

- [ ] Guardian add pannumbodhu OTP consent + timestamp store (DLT consent template kooda match)
- [ ] Plivo kitta "non-BFSI SOS service calls ku 080/022 OK ah, 160/1600 future la venuma" nu written confirmation
- [ ] TRAI direction updates ah 3 maasathuku oru thadava check

Sources: TRAI regulation text (Feb 2025), PIB press release, TRAI 1600-series direction (NCDEX circular), 1600-series guide (FreJun), Feb 2025 amendment summary (Rootle). Non-BFSI interpretation telecom lawyer kitta confirm pannunga.

7.7 India/US thaandi irukura guardians (UAE, UK, NRI)

Foreign guardians ku WhatsApp dhan main channel; IVR/SMS international ah costly + unreliable. WhatsApp ulagathula enga irundhalum ore Meta API la pogum, extra setup illa.

Channel	India number guardian	Foreign number guardian
WhatsApp	Yes, `devi_safety`	Yes, same template; Meta charge recipient country rate poruthu
IVR call	Yes, Plivo ₹0.38/min	Check: Plivo India account la international enable aagudha + rate confirm aagala (console la check). Illana Twilio/Telnyx ah international route ah use pannalam (destination rate)
SMS	Yes, DLT route	Check: International SMS DLT route la pogathu; provider international route + country rules (e.g. UAE sender ID registration) venum

Code impact: guardian number country code (`+91` illana) paathu routing: WhatsApp always; call/SMS international route enabled ah iruntha mattum. Ippo `whatsapp.service.js` 10-digit number ku `91` add pannudhu; foreign number full country code oda save pannanum.

Plivo note: Plivo India pages la India rate ku conflict iruku (₹0.38 / ₹0.40 / ₹0.60 per min vera vera pages la). Console la final rate check pannunga. (Plivo India pricing, Plivo country pricing help)

8. US compliance – A2P 10DLC & Toll-Free Verification

US numbers ku SMS anupanumna registration mandatory; illana carriers (AT&T, T-Mobile, Verizon) throttle or block pannuvaanga. DEVI guardians US la irundha mattum idhu venum.

Requirement	Enga apply	Enna pannanum
A2P 10DLC	US long-code (10-digit) number la irundhu application SMS	The Campaign Registry (TCR) la Brand + Campaign register (provider console vazhiya). 2021 la irundhu mandatory
Toll-Free Verification	US/Canada toll-free number la irundhu SMS	Use case submit pannanum. Free, mandatory, few days
Geo / dialing permissions	High-risk countries ku outbound call	Default ah off. Venum na explicit ah on pannanum (toll fraud prevention)

Brand/campaign fees provider console la kaatum; register panna munnadi current fee check pannunga.

9. WhatsApp – Meta Cloud API

DEVI already direct Meta Cloud API use pannudhu (`backend/src/services/whatsapp.service.js`, Graph API `v21.0`), so provider markup zero. Approved template: `devi_safety`.

9.1 Ippo namma setup

Item	Value
Endpoint	`https://graph.facebook.com/v21.0/{WHATSAPP_PHONE_NUMBER_ID}/messages`
Env variables	`WHATSAPP_PHONE_NUMBER_ID`, `WHATSAPP_ACCESS_TOKEN`, `ADMIN_WHATSAPP` (control room)

Item	Value
Template	`devi_safety`, language `en_US`, category Utility
Parameters	`name`, `location_link` (Google Maps link), `alert_time` (IST)
No guardian case	`ADMIN_WHATSAPP` control room ku pogum
Abuse limit	Unverified guardian number ku 1 hour la max 3 alerts

9.2 Business verification & template

1. business.facebook.com → Business Settings → Security Center la business verify pannunga (business licence / incorporation + business name & address match aagura document).
2. WhatsApp Manager la phone number add + display name approval.
3. Template **Utility** category la submit pannunga. Marketing category la potta SOS delay/block aagalam.
4. Verification free; Meta delivered message ku dhan charge pannum.

9.3 Watch out

- **Messaging limits:** new business account ku daily business-initiated message limit iruku. Verification + good quality rating apram dhan limit increase aagum. SOS scale aagura munnadi limit ah check pannunga.
- **Quality rating:** users block/report panna rating kuraiyum, limit um kuraiyum.
- **Delivery status:** Meta webhook la `sent`, `delivered`, `read`, `failed` status varum. Adha store panni dashboard la kaatunga.
- **Language:** ippo `en_US` mattum. Tamil/Hindi template thaniya approve pannalam.
- **Testing:** Meta API Setup page la test number + max 5 verified recipients (Section 4).

Action items (Meta): India WhatsApp account ah **Dec 31, 2026 kulla INR billing ku maathanum**, illana delivery disruption aagalam. Business Manager la payment method irukanum; Sep 30, 2026 kulla illatha accounts ku Oct 1 la irundhu service messages nikkalam. (Meta pricing)

10. Package / bundle / Enterprise tiers

Moonula yaarum fixed-price "all-in-one combo" package vikkala. Ellame pay-as-you-go: number rent, voice minute, SMS/MMS segment, WhatsApp message ellam thani thani bill. Difference: cheap self-serve tier la SMS/WhatsApp kidaikuma illa Enterprise la mattuma.

Provider	Self-serve la kidaikuradhu	Enterprise venum	Approx Enterprise cost
Twilio	Voice + SMS + MMS + WhatsApp + Numbers, day 1 la irundhu	Volume discount / dedicated support mattum	Sales kitta custom pricing
Plivo	Voice + Numbers + SIP Trunking + Audio Streaming + AI Agents; US & India mattum; $2,500/month cap	SMS, WhatsApp API, Verify API + US/India thaandi coverage	~$1,000/month la irundhu (third-party reports $750–$999; plivo.com/pricing la confirm)
Telnyx	Voice + SMS + MMS + Numbers + WhatsApp calling, day 1 la irundhu	Volume discount / priority support optional	Reported (unverified): Committed ~$500/mo, Growth ~$1,000/mo, Enterprise ~$5,000/mo

Takeaway: Twilio & Telnyx la Number + SMS + MMS + WhatsApp ellam pay-as-you-go la kidaikum. Plivo voice ku cheap, aana SMS/WhatsApp venumna Enterprise (~$1,000/month) ku poganum; appo combined use case ku Plivo dhan costly aagidum. Adhanala DEVI la Plivo = voice mattum.

11. Number vaangura process + Terms & Conditions

Moonu provider layum same 7 steps dhan.

11.1 Buying steps

1. **Sign up:** email vechu account. Telnyx ku business email venum (Gmail/Yahoo illa).
2. **KYC:** Telnyx mandatory document check. Twilio/Plivo full limits ku business details kekkum.
3. **Payment + balance:** card add panni balance top-up (number rent + expected usage).
4. **Search numbers:** country, area code, type (local / mobile / toll-free) vechu filter.
5. **Buy:** monthly rent udane charge aagum, auto-renew. Usage thaniya bill.
6. **Regulatory steps:** India, Brazil, EU mathiri countries ku address/business proof. US SMS ku A2P 10DLC or Toll-Free Verification.

7. **App kooda connect:** Voice/SMS/WhatsApp webhook URL ah namma server ku point pannunga (Section 13).

11.2 Accept panna vendiya rules

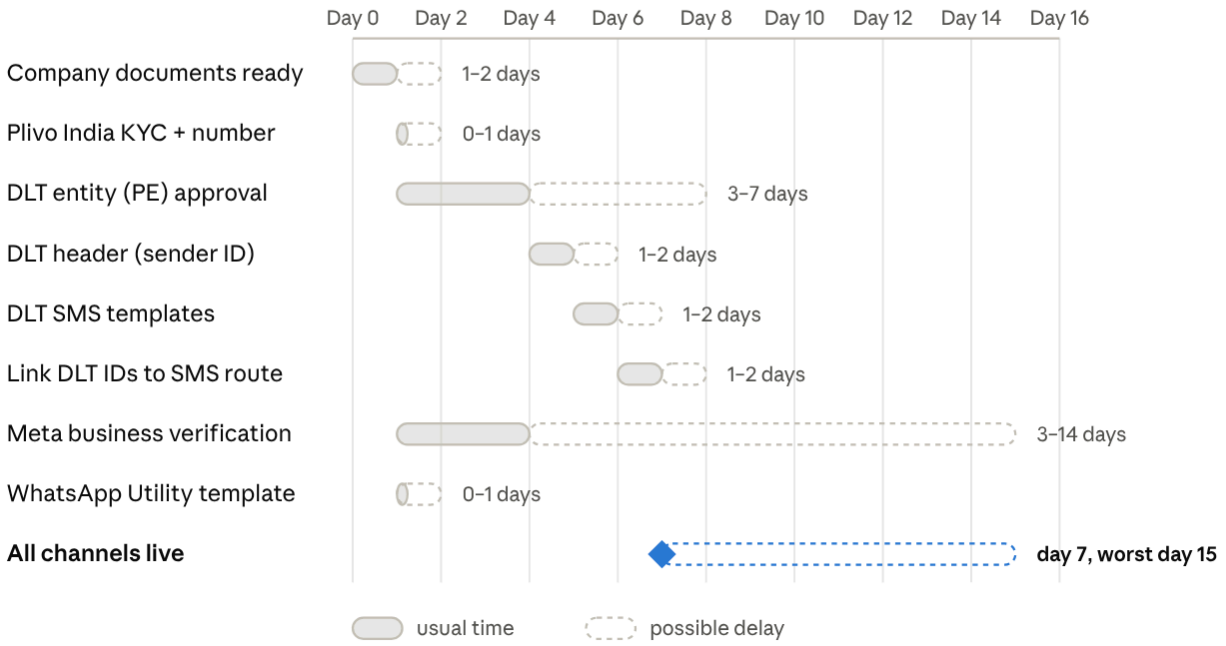
Requirement	Yaaruku	Yen
KYC / business verification	Telnyx (mandatory), Twilio & Plivo (higher limits/Enterprise)	Anti-fraud: spam, robocall, SIM-box fraud thadukka
A2P 10DLC	US long-code la irundhu application SMS	US carrier rule (2021 la irundhu); unregistered traffic throttle/block
Toll-Free Verification	US/Canada toll-free la SMS	Free, mandatory
Local presence documents	India, Brazil, sila EU/APAC numbers	Local regulator real presence proof kekkum
Geo / dialing permissions	High-risk/high-fraud destinations ku outbound	Default off, toll fraud thadukka
Acceptable Use Policy	Moonu perum	Consent illatha robocall, spam SMS, unsolicited WhatsApp ban. Violation panna account suspend

12. Certificates & approvals – process, templates, evlo days

Documents ready ah irundha **usual ah ~7 working days la, worst case ~15 working days (3 weeks) la** SMS + IVR + WhatsApp ellam live aagidum. Critical path DLT dhan: PE → header → templates onnu apram onnu. Plivo, Meta rendum parallel ah pogum.

Ellam sari ah pona 7 working days la SOS full live

Approx working days. DLT steps onnu apram onnu; Plivo + Meta parallel ah



Approvals timeline · approx working days

DLT steps sequence la pogum, so DLT ah day 1 laye start pannunga. Meta verification late aanalum WhatsApp 250 contacts/day limit oda munnadiye work aagum.

12.1 Ella approvals um oru paarvai la

Approval	Enga apply	Documents	Fee (approx)	Days (approx)	Kidaikuradhu
DLT Principal Entity (PE)	Oru operator DLT portal: Jio TrueConnect, Airtel, Vi, BSNL (oru idathula panna ellarukum share aagum)	Company PAN, GST certificate (or COI / Udyam), letterhead la Letter of Authorization, signatory ID + address proof, business email	~₹5,900 one-time (sila guides "₹5,900 + GST" nu solludhu; portal la check)	1–3 working days (sila case 7 varai)	PE ID
DLT Header (sender ID)	Same DLT portal	PE ID	Free	1–2 working days	6-letter header, e.g. `DEVIAP`
DLT content templates	Same DLT portal	Header + template text	Free	1–2 working days per template	Template ID per message

Approval	Enga apply	Documents	Fee (approx)	Days (approx)	Kidaikuradhu
DLT consent template	Same DLT portal (Service Explicit messages ku)	Consent text	Free	1–2 working days	Consent template ID
Plivo India number KYC	Plivo console → Phone Numbers → KYC (India-region account)	GST certificate, COI, or Udyam la onnu. First application ku authorized signatory sign + company seal. PAN mattum accept illa	KYC free; number ~₹200/month	080/022 number ~5 min (automated); 140/160 series thaniya SLA	India number for IVR
Meta business verification	business.facebook.com → Security Center	Legal business document (GST / COI / Udyam), business name + address match aaganum, website	Free	Usually few days, sometimes ~2 weeks	Verified business, 2,000/day limit
WhatsApp display name + template	WhatsApp Manager	Display name, template text, Utility category	Free (message ku dhan charge)	Minutes to 24 hours	Approved template
US A2P 10DLC (US guardians iruntha mattum)	Twilio/Telnyx console (TCR)	EIN, business details, website, sample messages, opt-in proof	Brand ~$4 (standard brand + vetting ~$46), campaign vetting ~$15, monthly campaign fee ~$1.50–$10	Brand minutes–days, campaign days–weeks	Registered campaign

Sources: Plivo India Number KYC, Meta messaging limits, DLT guide (QuickAuth, 2026), Infobip DLT docs, Bandwidth 10DLC fees (Feb 2026). DLT fees/timelines third-party guides la irundhu; operator portal la final check pannunga.

12.2 Step by step

A. DLT (SMS ku)

1. Documents ready: company PAN, GST certificate (or COI / Udyam), LOA (template 12.3), signatory Aadhaar/Passport.

2. Oru operator portal la (e.g. Jio TrueConnect) "Principal Entity" ah sign up, documents upload, fee pay → PE ID (1–3 days).
3. Header register: 6 letters, brand name kooda related, **Service/Transactional** category → 1–2 days.
4. Consent template + content templates (12.3) submit. Message la varra URL domain (e.g. `maps.google.com`) whitelist pannunga → 1–2 days.
5. Telemarketer (namma SMS provider) ah PE kooda bind pannunga.
6. PE ID, Header, Template IDs ah SMS provider account + namma `.env` la podunga. Test SMS anupi delivery check.

B. Plivo India number (IVR ku)

1. Plivo la **India data region** organization create (US region la Indian number kidaikathu; region apram maatha mudiyathu).
2. Phone Numbers → KYC → New Application → India.
3. GST / COI / Udyam la onnu upload. First time signatory sign + company seal.
4. Business type: **Direct Brand**.
5. Approve aana (080/022 ~5 min) Buy Phone Numbers → India → number rent → Voice webhook set.

C. Meta WhatsApp

1. Meta Business Manager → Security Center → Start verification → legal name, address, phone, website + document upload.
2. WhatsApp Manager la phone number add + display name ("DEVI") approval.
3. Template Utility category la submit (12.3). Verification mudiyaama kooda template approve aagum.
4. Verify aana 250 → 2,000 unique contacts/24h. Apram quality nalla irundha + 7 days la half limit use panna, automatic ah 10,000 → 100,000 → unlimited (6 hours la upgrade).

D. US 10DLC (US guardians iruntha mattum)

1. Provider console la Brand register (EIN, legal name, website).
2. Campaign register: use case, sample messages, opt-in method, privacy policy link.
3. Approve aana number ah campaign kooda link.

12.3 Ready templates

Letter of Authorization (DLT ku, company letterhead la)

```
[Company letterhead]
Date: ____

To,
The DLT Operator (Jio / Airtel / Vi / BSNL)

Subject: Letter of Authorization for DLT Principal Entity registration

We, [Company legal name] (PAN: ____, GSTIN: ____), authorize
[Name], [Designation], to register and manage our Principal Entity,
Headers, Content Templates and Consent Templates on your DLT platform
on behalf of the company.

Authorized person: Name ____  Mobile ____  Email ____  ID proof ____

For [Company legal name]
[Signature]  [Company seal]
[Director / Authorized Signatory name]
```

DLT SMS templates (Service Explicit category + consent; variable `{#var#}`; start la variable vekka koodathu, rendu variable pakkathu pakkathu vekka koodathu)

```
T1 SOS alert:
DEVI SOS: {#var#} needs urgent help. Location: https://maps.google.com/?q=
{#var#} Time: {#var#} - DEVI

T2 Guardian added:
DEVI: You are added as an emergency contact by {#var#}. You will get SOS
alerts with location from the DEVI app. - DEVI

T3 SOS closed:
DEVI: The SOS alert for {#var#} raised at {#var#} is now closed and marked
safe. - DEVI

Consent template:
I agree to receive emergency SOS alerts from the DEVI app for my contact.
```

Sila operator URL ku `{#url#}` variable type kekkum; SMS provider kitta confirm pannunga. Anupura text template kooda exact ah match aaganum (space, punctuation kooda).

IVR voice script (TTS ku; English + Tamil)

```
EN: This is an emergency alert from the DEVI app. {name} needs your help
right now.
    Press 1 if you are responding. Location link is on WhatsApp and SMS.

TA: இது DEVI செயலியின் அவசர அழைப்பு. {name} அவர்களுக்கு உடனடி உதவி
தேவை.
    நீங்கள் உதவிக்கு வருகிறீர்கள் என்றால் 1 ஐ அழுத்தவும்.
```

WhatsApp templates (Utility category)

```
Existing (approved): devi_safety, en_US
  params: name, location_link, alert_time

New (suggest): devi_safety_closed, en_US
  Update from DEVI: the SOS alert from {{name}} at {{alert_time}} is now
closed and marked safe.

New (suggest): devi_safety, ta (Tamil version of the same alert)
```

12.4 Reject aagura common reasons

Approval	Reason	Fix
DLT PE	Legal name PAN/GST la oru mathiri, form la vera mathiri	Ella documents la exact same legal name
DLT template	Live SMS text template la irundhu maaridichu	Code la template text ah constant ah vechu, variable mattum maathu
DLT template	Start la variable, back-to-back `{#var#}{#var#}`, undeclared URL	Fixed text la start, variables naduvula text, URL whitelist
Plivo KYC	Document expired / inactive, govt record kooda match aagala, blurry, sign + seal illa	Portal la irundhu fresh copy, signatory sign + seal
Meta verification	Business name/address Business Manager la oru mathiri, document la vera	Same legal name + address; website la business details
Meta template	Marketing words (offer, discount) → marketing category ku maathum	Pure alert/update text mattum
US 10DLC	Website la privacy policy / opt-in illa, sample messages match aagala	Privacy policy + opt-in proof + real sample messages

12.5 Presentation la varra questions – ready answers

Question	Answer
OSP licence venuma?	Venaam. DoT 2020–21 la remove pannitaanga; provider VNO authorisation la operate pannum
DLT illama SMS anupa mudiyuma?	Mudiyathu. Operator scrub panni drop pannidum
IVR call ku DLT venuma?	Plivo 080/022 number non-BFSI service/transactional calls ku allowed, KYC mattum podhum. 140/160 series venumna extra setup. TRAI rule maarina Plivo kitta confirm
DLT ovvoru operator layum thaniya pannanuma?	Venaam. Oru operator la register panna ellarukum share aagum
Meta verification mudiyaama WhatsApp anupa mudiyuma?	Mudiyum, 250 unique contacts/24h limit oda. Verify aana 2,000, apram auto scaling
Template reject aana?	Edit panni resubmit. DLT 1–2 days, Meta minutes–24h
One-time cost evlo?	DLT ~₹5,900; header/templates free; Plivo KYC free (number ~₹200/month); Meta free; US 10DLC (venumna) ~$4–$46 + $15 vetting + monthly fee
Total evlo naal?	Usual ~7 working days, worst ~15 (3 weeks), documents ready ah irundha
Yaar sign pannanum?	Director / authorized signatory + company seal
Individual ah (company illama) panna mudiyuma?	Plivo India number ku registered Indian business venum (GST / COI / Udyam). DLT ku business PAN/GST venum

13. DEVI project la implement panra steps

WhatsApp already ready; IVR + SMS dhan puthusa build pannanum. Provider maarinalum (Twilio / Plivo / Telnyx) concept same: REST call + webhook + XML. Keela Twilio SDK example.

13.1 Steps

1. **Account + keys:** sign up, KYC, API credentials copy. Code la hard-code panna koodathu, `.env` la mattum.
2. **Env variables add:** `IVR_ACCOUNT_SID`, `IVR_AUTH_TOKEN`, `IVR_FROM_NUMBER`, `BACKEND_PUBLIC_URL`, SMS provider keys + DLT `PE_ID` / `TEMPLATE_ID`.
3. **Number vaangu:** Voice capability ulla number (India ku Plivo India-region).
4. **SDK install:** `npm install twilio` (or `plivo`, `telnyx`) backend la.
5. **Public webhook URL:** production la HTTPS domain. Local testing ku ngrok mathiri tunnel.
6. **`backend/src/services/ivr.service.js`** create: call start + answer XML.
7. **`backend/src/routes/ivr.routes.js`** create: `/api/ivr/answer`, `/api/ivr/gather`, `/api/ivr/status`. Ella route layum signature validation.
8. **`backend/src/services/sms.service.js`** create: DLT approved template text mattum anupum.
9. **`sos.routes.js` la wire pannunga:** WhatsApp dispatch pakkathula SMS + first guardian call start.
10. **Call status store:** SOS alert kooda call attempts (guardian, status, acknowledged time) save panni dashboard la kaatunga.
11. **Multi-level menu venumna** oru menu screen ku oru webhook.

13.2 IVR service (sketch)

```
// backend/src/services/ivr.service.js
import twilio from 'twilio';

const client = twilio(process.env.IVR_ACCOUNT_SID,
process.env.IVR_AUTH_TOKEN);
const BASE = process.env.BACKEND_PUBLIC_URL;

export const IVRService = {
  // Guardian ku call start. Victim name URL la anupa koodathu (logs la PII).
  callGuardian(toNumber, sosId) {
    return client.calls.create({
      to: toNumber,
      from: process.env.IVR_FROM_NUMBER,
      url: `${BASE}/api/ivr/answer?sosId=${sosId}`,
      statusCallback: `${BASE}/api/ivr/status?sosId=${sosId}`,
      statusCallbackEvent: ['completed'], // CallStatus: completed / busy /
no-answer / failed
      machineDetection: 'Enable',         // voicemail ah "answered" nu count
panna koodathu
```

```
      timeout: 25,
    });
  },

  // VoiceResponse XML escape pannum, so name la & or < irundhalum break
aagathu
  answerTwiml(victimName, sosId) {
    const r = new twilio.twiml.VoiceResponse();
    const g = r.gather({ numDigits: 1, timeout: 10, action: `/api/ivr/gather?
sosId=${sosId}` });
    g.say({ voice: 'Polly.Aditi', language: 'en-IN' },
      `Emergency alert from DEVI. ${victimName} needs help now. Press 1 if
you are responding.`);
    r.say({ voice: 'Polly.Aditi', language: 'en-IN' }, 'No input received.
Please check WhatsApp and SMS.');
    return r.toString();
  },
};
```

13.3 Routes (sketch)

```
// backend/src/routes/ivr.routes.js
import express from 'express';
import twilio from 'twilio';

const router = express.Router();
const verify = twilio.webhook(process.env.IVR_AUTH_TOKEN); // fake POST block

router.post('/answer', verify, async (req, res) => {
  const alert = await getAlert(req.query.sosId); // name DB la irundhu
  res.type('text/xml').send(IVRService.answerTwiml(alert.victimName,
alert.id));
});

router.post('/gather', verify, async (req, res) => {
  if (req.body.Digits === '1') await markAcknowledged(req.query.sosId,
req.body.To);
  res.type('text/xml').send('<Response><Say>Thank you. Location link is on
WhatsApp and SMS.</Say></Response>');
});

router.post('/status', verify, async (req, res) => {
  const failed = ['busy', 'no-answer',
'failed'].includes(req.body.CallStatus)
    || req.body.AnsweredBy?.startsWith('machine');
```

```
  if (failed && !(await isAcknowledged(req.query.sosId))) await
callNextGuardian(req.query.sosId);
  res.sendStatus(204);
});
```

`getAlert`, `markAcknowledged`, `isAcknowledged`, `callNextGuardian` namma DB/SOS logic la ezhudhanum. Proxy pinnadi run aana signature check ku `BACKEND_PUBLIC_URL` exact ah match aaganum.

13.4 SMS (India)

SMS text **DLT template kooda exact match** aaganum; variable mattum maarum. Example template: `DEVI SOS: {#var#} needs help. Location: {#var#} - DEVI` (DLT la register panna text). Free text anupuna operator drop pannum.

13.5 Provider down aana backup

Plivo SLA 99.95% (Plivo infosheet) = maasathuku ~22 nimisham varai down aagalam; SOS ku backup route venum. WhatsApp (Meta) thaniya provider, so voice down aanalum WhatsApp pogum.

Channel	Primary	Backup	Switch eppo
Voice IVR	Plivo India	Twilio (pay-as-you-go, minimum illa; costly + international CLI) or Exotel (plan commitment)	Plivo API error / 5 sec timeout / carrier failure status
SMS	MSG91 (DLT)	Gupshup (same PE + templates, telemarketer binding rendukum)	Send API fail or DLR failed
WhatsApp	Meta Cloud API	SMS + call already parallel ah pogudhu	Meta error code

Code: `IVRService` ah provider-agnostic interface ah ezhudhunga (`callGuardian`, `answerXml`); `IVR_PROVIDER_PRIMARY` / `IVR_PROVIDER_BACKUP` env. Primary 2 thadava fail aana circuit breaker open panni few minutes backup ku route pannunga. Backup provider ku separate number + KYC + balance munnadiye ready ah vechukonga.

Sources: Plivo infrastructure infosheet (99.95% SLA). Sila Plivo marketing pages 99.99% solludhu; contract SLA sales kitta confirm.

13.6 Development effort (estimate)

Oru backend + oru Flutter developer ku ~3 weeks (~15.5–16.5 dev days); approvals (7–15 working days) kooda parallel ah pogum. Idhu namma code base paathu potta estimate; team velocity poruthu maarum.

Task	Owner	Dev days
Plivo account, number, env setup	Backend	0.5
`ivr.service.js` + `ivr.routes.js` (answer, gather, status) + signature validation	Backend	2
Guardian call sequence (next guardian, control room) + call attempts DB table	Backend	2
Inbound helpline route	Backend	1
`sms.service.js` + DLT templates	Backend	1
`sos.routes.js` wiring + dashboard la call/SMS status	Backend + Web	2
Guardian OTP consent + "save DEVI number" prompt + 112 button	Flutter	2
Provider backup (interface + circuit breaker)	Backend	2
Monitoring + alerts	Backend	1
E2E testing + fixes	Both	2–3
Total		**~15.5–16.5**

Provider backup (2 days) ah phase 2 ku thalli vechaa first release ~2.5 weeks la ready.

14. Security, abuse & privacy

IVR webhook open ah irundha yaar venaalum fake ah SOS ah "acknowledged" panna mudiyum; adhu dhan first fix.

Risk	Enna aagum	Fix
Webhook spoofing	Fake POST `/api/ivr/gather` la SOS acknowledged aagidum	Provider signature validation (Twilio `X-Twilio-Signature`, Plivo/Telnyx signature headers) ella IVR route layum

Risk	Enna aagum	Fix
XML injection	User name la `&` or `<` irundha TwiML break aagum	SDK builder (`VoiceResponse`) use pannunga, illana escape
Cost abuse / toll fraud	Fake SOS spam panna call/SMS bill ezhum	Existing `sosTriggerLimiter` + recipient limit (3/hour unverified) IVR + SMS kum apply. Geo permissions India (+ venum na) mattum on. Low-balance alert
PII in logs	URL query la name/number log aagum	URL la `sosId` mattum. Logs la phone mask (already `maskPhone` iruku)
Secrets leak	Account hijack, bill	Keys `.env` la mattum, rotate pannalam, repo la commit panna koodathu
Privacy (DPDP Act 2023)	Guardian number, location, call recording personal data	Guardian consent record, recording venaamna off, retention period fix pannunga
Voicemail false ack	Guardian edukkalai aana "answered" nu theriyum	AMD + DTMF "1" vandha mattum ack

14.1 DPDP Rules 2025 – DEVI ku enna venum

DPDP Rules 13 Nov 2025 la notify aachu; main duties (notice, consent, security, retention, breach) 13 May 2027 la irundhu kattayam. Ippove build panna later rework illa.

Rule	Enna solludhu	DEVI la
Notice (Rule 3)	Data collect pannumbodhu enna data, purpose, rights nu clear notice	User location + guardian numbers ku notice screen
Consent	Clear, specific consent; withdraw panna easy ah irukanum	User consent + guardian OTP consent; app la remove/withdraw option
Consent Managers	13 Nov 2026 la irundhu	Ippo theva illa
Breach (Rule 7)	Board + affected users ku "without delay"; detailed report Board ku 72 hours la	Incident runbook + contact list ready

Rule	Enna solludhu	DEVI la
Logs retention (Rule 8(3))	Personal data, traffic data, processing logs minimum 1 year	SOS, call, SMS, WhatsApp logs 1 year keep; apram delete
3-year erasure (Rule 8(1))	2 crore+ users e-commerce/social media, 50 lakh+ gaming ku mattum	DEVI ku apply aagathu (ippo)
Security	Reasonable safeguards (encryption, access control, logging)	Already security audit la cover (see security doc)

Sources: DPDP Rules 2025 (Wikipedia), DPDP timeline (grc-flow), DPDP compliance timeline (ConsentOS), DSCI FAQ on retention & erasure, Mondaq on retention. Rule text MeitY gazette la verify pannunga.

14.2 Call recording consent & disclaimer

India la call recording ku clear statute illa; safe practice = record panradhu nu munnadiye solradhu, illana record pannama irukradhu. DEVI SOS call ku recording theva illa (DTMF "1" + status log podhum), so **default recording OFF** recommend.

Option	Eppo	Enna pannanum
Recording OFF (recommended)	SOS outbound + inbound	Call status, DTMF, timestamps mattum log
Recording ON	Control room ku connect aagura calls (dispute/audit ku)	Start la "This call is recorded for safety and quality" announcement; purpose + retention (e.g. 90 days) privacy policy la; access control

App disclaimer (privacy policy / T&C la): "DEVI emergency calls, SMS and WhatsApp alerts are sent to the contacts you add. DEVI does not replace 112 or police. Delivery depends on telecom and internet networks."

Sources: Call recording compliance India (Exotel), Telephone call recording laws (Wikipedia). Final wording ku lawyer review.

15. Testing & go-live checklist

Verified test numbers la ellam pass aana apram dhan real guardians ku on pannanum.

Testing

- [ ] Trial account la own verified number ku IVR call varudha
- [ ] "1" press panna dashboard la acknowledged aagudha
- [ ] Edukkama vittaa / busy / voicemail → adutha guardian ku pogudha
- [ ] Ellarum fail → control room (`ADMIN_WHATSAPP`) ku pogudha
- [ ] Fake webhook POST (signature illama) reject aagudha
- [ ] Name la special characters (`&` , `<`) irundhalum call work aagudha
- [ ] WhatsApp `devi_safety` + SMS same time la varudha
- [ ] Internet illatha phone la SMS varudha

Go-live

- [ ] DLT PE + Header + SMS template + consent template approved
- [ ] India-region number (Plivo) KYC mudinjuthu
- [ ] Meta business verification + Utility template approved
- [ ] US guardians iruntha A2P 10DLC / Toll-Free Verification
- [ ] Dial pannura countries ku geo permission on
- [ ] Production HTTPS webhook URL provider console la set
- [ ] Auto-recharge + low-balance alert on
- [ ] Call/SMS/WhatsApp failure monitoring + delivery status logging

Monitoring & alerts

Ovvoru SOS kum "guardian kitta alert reach aachaa" nu dashboard la theriyanum; fail aana udane control room ku alert pogaanum.

Signal	Source	Store / show
Call status	Plivo status webhook (`completed` , `busy` , `no-answer` , `failed` , machine)	Call attempts table, dashboard
DTMF ack	`/api/ivr/gather`	SOS "acknowledged" + time
SMS delivery (DLR)	SMS provider DLR webhook (`delivered` , `failed` , DLT scrub error)	SMS log
WhatsApp status	Meta webhook (`sent` , `delivered` , `read` , `failed`)	WhatsApp log
Provider balance	Plivo / MSG91 balance API	Daily check

Alerts:

- [ ] SOS 2 nimisham la yaarum acknowledge pannala → control room alert
- [ ] Call failure rate > 20% (last 15 min) → backup provider + dev alert
- [ ] SMS DLT scrub failures → template mismatch alert
- [ ] WhatsApp `failed` spike or quality rating drop → alert
- [ ] Provider balance < 2 days usage → recharge alert

Thresholds (2 min, 20%) starting values; real data paathu tune pannunga.

16. Final comparison & recommendation

DEVI ku: Plivo (India voice) + India DLT SMS + direct Meta WhatsApp. Global guardians adhigama aana Twilio voice add pannalam.

Criteria	Twilio	Plivo	Telnyx
Global coverage	Best (100+)	Good (65+, self-serve US/India mattum)	Moderate (50+), US/UK/EU strong
Pricing	Premium (highest)	Cheapest overall	Low, transparent pay-as-you-go
India voice	Costly, Indian number illa, international CLI	Cheapest, Indian number iruku	India rate publish aagala, 4 cities numbers
Free trial	Decent bundle, card illa	$10, card illa, SMS/WhatsApp locked	$25 AI pretrial, business email + KYC

Criteria	Twilio	Plivo	Telnyx
IVR tooling	Best (Studio + TwiML)	Good (PlivoXML + basic Flow)	Good (TeXML + Call Control API + AI voice)
SMS + WhatsApp self-serve la	Yes	No (Enterprise)	Yes
Best for	Enterprise, widest reach, fast no-code	Cost-conscious, India-heavy traffic	Full API control, lowest latency

Moonum webhook + markup concept la same, so later provider maathuradhu easy. Adhanala `IVRService` ah provider-agnostic ah ezhudhunga.

17. Sources & disclaimer

Intha guide `docs/Twilio_Plivo_Telnyx_Full_Guide.pdf` (October 2026) + DEVI backend code (`whatsapp.service.js`, `sos.routes.js`) base pannitu ezhudhinadhu. PDF oda sources (accessed October 2026):

- Twilio pricing, Twilio India voice guidelines, Twilio docs
- Plivo pricing, Plivo India voice pricing, Plivo rent India numbers
- Telnyx pricing, Telnyx developers
- TRAI (TCCCPR regulation text)
- Third-party trackers: cloudtalk.io, dialsheet.io, krispcall.com, smscountry.com, frejun.com

PDF la illatha sila points (Meta free tier SOS ku apply aagathu, India utility ~₹0.12, DLT fee ~₹5,900, WhatsApp messaging limits, DPDP Act) approximate; official page la verify pannunga.

Disclaimer: CPaaS pricing um India telecom rules um adikadi maarum. Number vaanga, campaign launch panna munnadi provider / TRAI / Meta official site la current rate & compliance check pannunga.